

can be cracked easily. However, that's true only if the "algorithm" is known by the person trying to crack the cipher (someone who's not the intended recipient). In this case we won't be increasing the block size but altering the key size so that the strength of the cipher increases. You could also increase the block size to further increase the strength, but we're trying to keep things simple to understand.

We're going to increase the key size using a very simple technique – we take the original key (32-bit) and split it into four parts. Remember, each bit sequence of 4 bits corresponds to one hexadecimal number. We will use that here instead of dealing with bits directly to make it easy to read. Our key when represented in hexadecimal would be a sequence of 8 hexadecimal numbers. For simplicity, our key will be "02468ACE".

We're going to derive a 128-bit key from the original 32-bit key by working with chunks of 8 bits from the original key using the following table:

**TABLE 2: GENERATING THE 128-BIT KEY FROM THE 32-BIT KEY**

|                              | Part 1 | Part 2 | Part 3 | Part 4 |
|------------------------------|--------|--------|--------|--------|
| 0th Iteration (original key) | 02     | 46     | 8A     | CE     |
| 1st Iteration                | CE     | 02     | 46     | 8A     |
| 2nd Iteration                | 8A     | CE     | 02     | 46     |
| 3rd Iteration                | 46     | 8A     | CE     | 02     |

If you notice, the pattern is actually simple: we rotate the chunk's positions by one and we do this three times. We then write them in sequence making the derived key:

02468ACECE02468A8ACE0246468ACE02

We'll use this key to run the main algorithm. Before we proceed, there are a few things you need to note:

- ▶ The algorithm we've used to derive the 128-bit key from the original 32-bit key is very weak and can't be used for any real secure communication.
- ▶ There are better ways to generate a 128-bit key from a 32-bit key. For example, you can do bit shuffling rather than shuffling blocks of 8-bits. By nesting rotations (rotating bits of chunks, as well as rotating the chunks themselves) the generated sequence would make the procedure of tracking the 32-bit key from the 128-bit key a bit more difficult.
- ▶ We could have chosen a 128-bit key in the very beginning, it would have served as well, but then, we would have to send the 128-bit key – thus making it impractical. We do the 32-bit to 128-bit transformation so that